

Yunshan Duan

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EMPLOYMENT

Postdoctoral Fellow, Johns Hopkins University

Aug 2025 - Present

Department of Applied Mathematics and Statistics

Advisor: Dr. Yanxun Xu

EDUCATION

University of Texas at Austin

Austin

Ph.D. in Statistics

Aug 2025

Advisor: Dr. Peter Müller

Dissertation: Bayesian nonparametric inference in bioinformatics applications

Fudan University

Shanghai

B.S. in Mathematics and Applied Mathematics

Jun 2021

RESEARCH INTERESTS

My research develops **statistical machine learning** methods for complex biomedical data through interdisciplinary work connecting statistics, machine learning, and biomedical science.

- **Methodology:** representation learning; Bayesian nonparametrics; dependent random partitions; Gaussian processes; generalized Bayes; Bayesian adaptive clinical trial design; sequential decision-making.
- **Applications:** electronic health records (EHR); cancer genomics; single-cell and spatial transcriptomics; early-phase clinical trials; surrogate endpoints; HIV; mental health and cognition.

PREPRINTS & PUBLICATIONS

- **Duan, Y., Korpela, E., & Xu, Y. (2026).** Self-Supervised Representation Learning for Longitudinal Polypharmacy Patterns. arXiv preprint arXiv:2608.06743. (*Under Review*).
- **Duan, Y., & Williamson, S. (2026).** Self-Supervised Learning with Gaussian Processes. arXiv preprint arXiv:2512.09322. (*Under Review*).
- **Duan, Y., Guo, S., Yan, H., Wang, W., Müller, P. (2025).** Spatially aligned random partition model on spatial transcriptomics data. *Bayesian Analysis*, Advance Publication 1-26, 2025.
- **Duan, Y., Guo, S., Wang, W., & Müller, P. (2025).** Immune profiling among colorectal cancer subtypes using dependent mixture models. *Journal of the American Statistical Association*, 1-23. (**Winner of 2024 ASA-SBSS student paper competition**)
- **Duan, Y., & Parast, L. (2024).** Flexible evaluation of surrogate markers with Bayesian model averaging. *Statistics in Medicine*, 43(4), 774-792.
- **Duan, Y., Yuan, S., Ji, Y., & Müller, P. (2024).** A unified decision framework for phase I dose-finding designs. *Statistics in Biosciences*, 16(1), 69-85.
- **Tec, M., Duan, Y., & Müller, P. (2023).** A comparative tutorial of Bayesian sequential design and reinforcement

learning. *The American Statistician*, 77(2), 223-233.

- **Duan, Y.**, Müller, P., & Ji, Y. (2023). Comment: Response-adaptive randomization in clinical trials: From myths to practical considerations. *Statistical Science*, 38(2), 212-215.
- Müller, P., **Duan, Y.**, & Garcia Tec, M. (2022). Simulation based sequential design. *Pharmaceutical Statistics*, 21(4), 729-739.
- **Duan, Y.**, Wang, S. J., & Ji, Y. (2021). Hi3+3: A model-assisted dose-finding design borrowing historical data. *Contemporary Clinical Trials*, 106437.

WORKING PAPERS

- **Duan, Y.**, Guo, X., Zhong, W. Synthesis of multiple endpoints by Bayesian multi-state modeling in oncology trials. (*Manuscript in preparation*).
- **Duan, Y.**, Korpela, E., & Xu, Y. Regimen aware analysis of CNS-active medication effects on cognition in people with HIV. (*Manuscript in preparation*).

BOOKS

- Schwob, M., **Duan, Y.**, Cantoni, B., Flores-López, B., & Walker S. (2025). *Exercises in statistical reasoning*, Chapman & Hall/CRC.
- Zhuo, J., **Duan, Y.**, & Jiang X. (2022). *MATLAB operation research*, Tsinghua University Press.

AWARDS & HONORS

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|--|------------------------------|
| • University Graduate Continuing Fellowship, UT Austin | 2024-2025 |
| • Second Place Poster Award of 2025 Best of Statistical Science Workshop | Apr 2025 |
| • Winner of 2024 ASA-SBSS student paper competition | Aug 2024 |
| • Winner of the 5 th Stat4Onc Annual Symposium poster competition | May 2022 |
| • Travel Funding Award | May 2022; May 2023 |
| • Freshman scholarship and Scholarship of Fudan University | Dec 2017; Oct 2018; Oct 2019 |

TEACHING EXPERIENCE

Johns Hopkins University

Guest Lecturer

- EN.553.734 Generative AI and Agentic Systems for Clinical Data Science and Biopharmaceutical Statistics (*graduate level*), Fall 2026.

University of Texas at Austin

Teaching Assistant

- SDS 320E Elements of Statistics, Fall 2021, Spring 2024.
- SDS 322E Elements of Data Science, Spring 2022.

INVITED TALKS

- Frontiers of Bayesian Inference and Data Science, Sep 2024, Oaxaca, Mexico.
Self-supervised learning with Gaussian Processes
- Interpretable Inference via Principled BNP Approaches in Biomedical Research and Beyond, Jul 2024, Singapore.

Spatially aligned random partitions for tumor-immune interaction

- 2023 WNAR/IMS Annual Meeting, June 2023, Anchorage, AK.

Flexible evaluation of surrogate markers with Bayesian Model Averaging

CONTRIBUTED TALKS & POSTER PRESENTATIONS

- Joint Statistical Meeting, Aug 2026, Boston, MA.

Self-supervised representation learning for longitudinal polypharmacy patterns

- 2025 Best of Statistical Science Workshop (Poster Session), Apr 2025, College Station, TX.

Spatially aligned random partitions on spatial transcriptomics data

- Joint Statistical Meeting (Student Paper Competition), Aug 2024, Portland, OR.

Immune profiling among Colorectal Cancer subtypes using Dependent Mixture Model

- 2023 Leading Edge of Cancer Research Symposium (Poster Session), Nov 2023, Houston, TX.

Immune profiling among Colorectal Cancer subtypes using Dependent Mixture Model

- Women in Machine Learning (Poster Session), Nov 2022, New Orleans, LA.

Spatial clustering with random partitions on ovarian cancer data

- Conference on advances in Data Science (Poster Session), Oct 2022, College Station, TX.

Immune profiling among Colorectal Cancer subtypes using Dependent Mixture Model

- New England Statistical Society Symposium, May 2022, Storrs, CT.

A Unified Decision Framework for Phase I Dose-Finding Designs

- Stat4Onc symposium (Poster Session), May 2022, online.

A Unified Decision Framework for Phase I Dose-Finding Designs

- Society for Clinical Trials Annual Meeting, April 2022, online.

A Unified Decision Framework for Phase I Dose-Finding Designs

SOFTWARE

- **SARP**: R implementation of spatially aligned random partition model on spatial transcriptomics data.
<https://github.com/YunshanDYS/SARP>
- **SASC**: R implementation of dependent mixture models for immune profiling among colorectal cancer subtypes.
<https://github.com/YunshanDYS/SASCcode>
- **SurrogateBMA**: R package for flexible evaluation of surrogate markers with Bayesian model averaging.
<https://cran.r-project.org/web/packages/SurrogateBMA/index.html>

PROFESSIONAL SERVICE

Conference Service

- Organizer, ENAR 2027 Spring Meeting Invited Session, “*Statistical Representation Learning for Biomedical Discovery from Complex Health Data*”.
- Scientific Committee for the Bayesian Young Statisticians Meeting (BAYSM) 2025.
- Organizer, 2023 WNAR/IMS Meeting Invited Session, “*Novel methods to identify & use surrogate markers in clinical trials*”.

Journal reviewer

- Biometrics, Statistical Science, Biostatistics, The New England Journal of Statistics in Data Science, BMC Research Notes, Journal of Classification.

INDUSTRY & RESEARCH EXPERIENCE

MD Anderson Cancer Center

June 2025 – July 2025

Research Intern

Mentor: Dr. Wenyi Wang

- Developed Bayesian methods for spatial transcriptomics with applications to characterizing spatial patterns, cellular heterogeneity and interactions in the tumor microenvironment.
- Collaborated with biomedical researchers on the analysis and interpretation of spatial omics data.

Pfizer

May 2023 – Aug 2023

Biostatistics Intern

Mentor: Dr. Wei Zhong, Dr. Xiaohan Guo

- Investigated synthesis of multiple endpoints in oncology trials using Bayesian multi-state models.
- Developed a Bayesian model allowing for joint inference of overall survival, response, and progression-free survival.

Cytel

Aug 2020 – Aug 2021

Biostatistics Intern

Mentor: Dr. Yuan Ji

- Developed statistical engines using C++ and R. Implemented trial designs, debugged, and validated results.
- Developed R package/ R shiny page for novel Bayesian clinical trial designs.
- Conducted clinical trial consulting. Provided clinical trials conducting suggestions, simulation results, edited protocols.